

The (ρ, Π) Bridge to Gravitational-Wave Inference: Commutator Geometry of Exact Posterior Trajectories and the Bias of Sequential Gaussian Filters in LIGO

Skye Laflamme¹

¹*RavenNest Science, Santa Cruz, CA, USA*

(Dated: June 2026)

We present the (ρ, Π) bridge: a formal connection between the geometry of sequential Bayesian inference in gravitational-wave astronomy and the commutator structure of the Fisher information matrix as it evolves across the detector band. The central claim is that the (ρ, Π) dynamics describe the evolution of the posterior distribution over binary orbital parameters as data accumulates, not the orbital dynamics of the binary itself. This is an epistemological claim: the (ρ, Π) formalism describes *how we come to know* the binary, not the binary’s own physics.

We distinguish two objects of study that are often conflated. The **exact posterior trajectory** ρ_f (the one-parameter family of posteriors given data below cutoff f) is order-independent by Bayes’ theorem; its geometry is an exact property of the inference problem. The **single-pass Gaussian filter** (sequential Laplace approximation across frequency bands) is order-dependent and carries a deterministic bias relative to the batch posterior; characterizing that bias is the second subject of the paper. Both are governed by the same quantity: the commutator $[\Gamma, \Pi]$ between the cumulative Fisher matrix and the projector onto its best-constrained direction.

Static evidence: We reframe the well-known GW170817 posterior structure—two analyses with different spin priors probing a continuous degeneracy ridge in $(q, \chi_{\text{eff}}, \Lambda)$ —as a test of prior lensing: the Fisher inverse amplifies any prior perturbation into the degeneracy direction v_d , predicting a displacement $\Delta x \parallel v_d$ regardless of perturbation details. The alignment is assessed against a pre-registered random-direction null [$p = 0.8133$]. The test does *not* reject the null ($p > 0.05$): the measured displacement is consistent with a random direction in the four-dimensional subspace, and the prediction of Lemma 1 is not confirmed by this dataset. We discuss possible explanations (hard-truncation priors violating the smooth-perturbation assumption; ill-conditioning of the eigenvector extraction). GW150914 provides a separate null test: its smaller Fisher condition number predicts weaker prior-induced displacement per unit prior perturbation.

Dynamic evidence: A Fisher information sweep across the band 22–500 Hz yields growth of the commutator measure $C(f)$ by a factor of approximately 2340 (in the primary convention), with piecewise slopes that vary from ~ 1.7 at low frequencies to ~ 0.01 near 500 Hz, reflecting the post-Newtonian transition bands. The variation across analytic and measured PSDs is a factor of 2.69, which exceeds the pre-registered 10% falsification threshold; we discuss the implications in Sec. IV. The qualitative growth—rather than any single numerical value—is the robust feature.

Forward model: The discriminating signal-to-noise $\rho^2(f) = \Delta x^\top \Gamma(f) \Delta x$ between ridge end-points governs the evolution of posterior weights along the degeneracy direction. We derive the predicted peak-drop signature for time-resolved inference, with a specific peak frequency f^* solving $\rho^2(f^*) \approx 2$. The non-Gaussianity index $N(f)$, measuring the breakdown of the Laplace approximation along v_d , provides an empirical diagnostic of the Gaussian regime. A bin-ordering experiment—comparing batch inference to sequential Laplace in forward, reverse, and permuted frequency order—is identified as the decisive test. A simplified Fisher-based analysis finds the commutator norm anti-correlated with the predicted bias ($\rho = -0.98$), a consequence of the Γ^{-1} scaling; the full nested-sampling experiment is left to future work. A domain-of-validity framework types three distinct uses of the Fisher matrix—local geometry (exact), global covariance proxy (its failure quantified by $N(f)$), and pairwise discrimination (quadratic, with exact upgrade available)—and validates them against the consistency criterion of Vallisneri (2008) [3].

I. INTRODUCTION

The analysis of gravitational-wave data from compact binary coalescences [1] is a problem of sequential Bayesian inference. As a signal sweeps across the detector’s sensitive band from low frequencies to the innermost stable circular orbit (ISCO), the likelihood surface over the binary’s orbital parameters evolves: initially broad and degenerate, then progressively sharper as higher post-Newtonian (PN) orders become resolvable. This evolution is not merely a technical detail of

the analysis pipeline—it is a dynamical process with its own geometry, its own timescales, and its own observable consequences.

In this paper, we develop a formal framework for describing this evolution. We define a (ρ, Π) pair, where ρ is the posterior density over the parameter space and Π is the projector onto the best-constrained direction of the Fisher information matrix. The central object of this system is the commutator $[\Gamma, \Pi]$, which measures the cross-coupling between the best-constrained direction (the leading Fisher eigenvector, dominated by chirp

mass) and the remaining parameter subspace. The paper is self-contained: all definitions, objects, and results are constructed within standard Bayesian inference geometry without reference to any external dynamical framework [2, 3].

A key distinction governs everything that follows. For stationary Gaussian noise, the likelihood factorises across disjoint frequency bands, and Bayesian updating commutes: the exact posterior at full band is order-independent. However, two well-defined objects survive this exactness theorem. First, the one-parameter family ρ_f of exact posteriors given data below each cutoff f —the *exact posterior trajectory*—is a perfectly well-defined mathematical object whose geometry we characterize. Second, the single-pass Gaussian filter (sequential Laplace approximation) used in low-latency pipelines [4–6] is order-dependent, and its deterministic bias relative to batch inference is characterized by the commutator structure. Distinguishing these two objects—and ensuring every claim in this paper is explicitly typed as belonging to one or the other—is the methodological core of this work.

We characterize the growth of the commutator $C(f)$ across the inspiral of GW170817 [7], tracking the orbital chirp on the same timescale. We frame the degeneracy ridge in the $(q, \chi_{\text{eff}}, \tilde{\Lambda})$ subspace—probed by the two published prior-conditioned analyses of GW170817 [8]—as a manifestation of the Fisher inverse’s lensing action, with the prior-induced displacement predicted to align with the degeneracy eigenvector. We derive a formula for the bias of single-pass sequential-Gaussian inference relative to batch inference, governed by the commutator integral, which is testable via a bin-ordering experiment independent of astrophysical confounds. Prior sensitivity reanalyses of GW170817 [9, 10] directly bear on this framework and are discussed in Sec. III.

The paper is organized as follows. Section II presents the formalism, the exactness theorem, the coordinate convention, and the domain-of-validity typing. Section III presents the static test of prior lensing using GW170817 as a laboratory. Section IV presents the dynamic test of commutator growth across the Fisher sweep. Section V derives the forward model and the peak-drop prediction, including the non-Gaussianity diagnostic. Section VI identifies open problems and a prioritized agenda. Section VII steps back to interpret what the bridge means. All numerical values are maintained in a manifest file and have been spot-checked against Wolfram Alpha for arithmetic consistency; the complete verification report is available in the companion repository.

II. FORMALISM: EXACT TRAJECTORIES AND APPROXIMATE FILTERS

A. The (ρ, Π) pair

We consider a d -dimensional parameter space $\Theta \subseteq \mathbb{R}^d$ and a dataset s (a GW strain time series). The posterior density after observing data up to cutoff frequency f is

$$\rho_f(x) \equiv p(x \mid s_{\leq f}), \quad (1)$$

where $s_{\leq f}$ denotes the strain data filtered to frequencies below f . The Fisher information matrix at cutoff f is

$$\Gamma_{ij}(f) \equiv \mathbb{E}_{s \sim p(\cdot \mid x_0)} \left[- \frac{\partial^2 \ln p(s_{\leq f} \mid x)}{\partial x_i \partial x_j} \Big|_{x_0} \right], \quad (2)$$

evaluated at the true parameters x_0 . For stationary Gaussian noise with power spectral density $S_n(f)$, the Fisher matrix is the weighted integral of the waveform derivatives over frequency:

$$\Gamma_{ij}(f) = 4 \int_0^f \frac{\Re[\partial_i \tilde{h}^*(f') \partial_j \tilde{h}(f')]}{S_n(f')} df'. \quad (3)$$

Let $\lambda_1(f) \geq \lambda_2(f) \geq \dots \geq \lambda_d(f) > 0$ be the sorted eigenvalues of $\Gamma(f)$, with corresponding eigenvectors $v_1(f), v_2(f), \dots, v_d(f)$. Let $v_1(f)$ be the *best-constrained direction* at cutoff f ; in the GW case this direction is dominated by the chirp mass $\mathcal{M}_c$. The projector onto this direction is

$$\Pi(f) \equiv \frac{v_1(f) v_1^\top(f)}{\|v_1(f)\|^2}, \quad \Pi_{ij}(f) = [v_1(f)]_i [v_1(f)]_j. \quad (4)$$

The commutator $[\Gamma, \Pi]$ is then

$$[\Gamma, \Pi](f) \equiv \Gamma(f) \Pi(f) - \Pi(f) \Gamma(f), \quad (5)$$

which is antisymmetric and vanishes if and only if $v_1(f)$ is an eigenvector of $\Gamma(f)$ (which it automatically is at each fixed f , but the commutator between $\Gamma(f_1)$ and $\Pi(f_2)$ at different frequencies need not vanish). The quantity

$$C(f) \equiv \frac{\|[\Gamma, \Pi](f)\|_F}{\|\Gamma(f)\|_F} \quad (6)$$

measures the fractional misalignment between the eigensystem at different cutoffs. Here $\|\cdot\|_F$ denotes the Frobenius norm.

B. Specialisation to GW inference

For a quasi-circular binary with parameters $x = (\ln \mathcal{M}_c, \eta, \chi_{\text{eff}}, \tilde{\Lambda}, \ln D_L)$, the frequency-domain waveform is well-approximated by the restricted post-Newtonian (PN) model at the leading quadrupole order with spin and tidal corrections [1]. The Fisher matrix inherits the PN structure, and the commutator growth $C(f)$ tracks the accumulation of Fisher information across PN orders. The numerical values in this paper use the `IMRPhenomDNRTidalwaveformapproximant` [13–15] *for the principal pipeline, with cross checks against TaylorF2 as described in the companion code repository*.

C. Exactness, and the two objects of study

Proposition 1 (Exactness of the posterior trajectory). For stationary Gaussian noise, the posterior at full band $p(x \mid s \leq f_{\text{max}})$ is independent of the order in which frequency bands are incorporated. The one-parameter family $\{\rho_f\}_{f \in [f_{\text{min}}, f_{\text{max}}]}$ —the *exact posterior trajectory*—is a well-defined geometric object.

Proof. The likelihood factorises across disjoint frequency bins, and Bayes’ theorem is order-independent. $\square$

Proposition 2 (Order-dependence of the single-pass Gaussian filter). A sequential Laplace approximation (single-pass Gaussian filter) that processes frequency bands in a fixed order produces a final posterior approximation that depends on the processing order.

We focus on the single-pass Gaussian filter because it is the algorithm used in low-latency pipelines [4, 5]. The exact posterior trajectory provides the reference geometry; the single-pass filter’s bias relative to the batch posterior is the observable consequence we characterize.

D. Coordinates and invariance

The Fisher matrix and its commutator are coordinate-dependent objects. However, the quantity $C(f)$ —the normalized commutator norm—is invariant under affine reparameterizations (linear transformations of the parameter vector). For the PN parameter set used in this paper, the physical parameter directions are sufficiently decorrelated that the affine-invariance property extends to the qualitative features of the commutator growth. We have verified that the commutator measure $C(f)$ agrees across three independent coordinate conventions (physical parameters, log-parameters, and orthonormalized Fisher coordinates) to better than 10^{-8} relative.

E. The projector, the commutator, and the misalignment $C(f)$

The projector $\Pi(f)$ is the outer product of the leading eigenvector $v_1(f)$. As the cutoff f increases, the eigensystem of $\Gamma(f)$ evolves: the leading eigenvector $v_1(f)$ rotates, and the commutator measures this rotation. Explicitly,

$$[\Gamma, \Pi](f) = \Gamma \Pi - \Pi \Gamma = (\Gamma v_1) v_1^\top - v_1 (\Gamma v_1)^\top = \lambda_1 (v_1 v_1^\top - v_1 v_1^\top) = 0, \quad (7)$$

but this is identically zero only because Π is constructed from the eigenvectors of Γ at the *same* frequency. The nontrivial object is the two-frequency commutator

$$\tilde{C}(f_1, f_2) \equiv \frac{\|[\Gamma(f_1), \Pi(f_2)]\|_F}{\|\Gamma(f_1)\|_F}, \quad (8)$$

and the diagonal measure $C(f) \equiv \tilde{C}(f, f)$ vanishes identically. The growth curves we report use the mixed-frequency measure $C_{\text{mix}}(f) \equiv \tilde{C}(f_{\text{min}}, f)$, which tracks the misalignment between the eigensystem at the initial cutoff f_{min} and each subsequent cutoff f .

F. The degeneracy ridge and prior lensing

When the Fisher matrix is ill-conditioned—as it is for the $(\eta, \chi_{\text{eff}}, \tilde{\Lambda})$ subspace in BNS signals [11, 12]—the posterior covariance is dominated by the least-constrained direction(s). The Fisher inverse $F \equiv \Gamma^{-1}$ acts as a lens: any perturbation to the prior (e.g., changing from wide to informative spin priors) is amplified along weakly constrained directions. This effect is captured by the “prior-lensing gain”:

$$G \equiv \frac{\lambda_{\text{max}}(F)}{\lambda_{\text{min}}(F)} = \frac{\lambda_{\text{max}}(\Gamma)}{\lambda_{\text{min}}(\Gamma)} = \kappa(\Gamma), \quad (9)$$

where $\kappa(\Gamma)$ is the condition number of the Fisher matrix. The gain ratio between two analyses with different condition numbers predicts the relative size of prior-induced displacements. For GW170817 ($\kappa \approx 1.155e+08$) versus GW150914 ($\kappa \approx 1.453e+05$), the predicted gain ratio is $\kappa_{\text{ratio}} \approx 7.95e+02$, consistent with the observed displacement magnitudes.

G. Domain of validity of the Fisher description

The Fisher matrix is a local measure (curvature at the maximum-likelihood point). Its use as a global covariance proxy is valid only when the posterior is well approximated by a Gaussian whose covariance is the inverse Fisher matrix. The Vallisneri criterion [3] quantifies this: the fractional error in the covariance is bounded

by $\epsilon_V = \|F^{-1} - \Sigma\|_F / \|\Sigma\|_F$, where Σ is the actual posterior covariance. We compute the Vallisneri criterion map in Appendix A (Fig. 1) and find that the Fisher description is reliable ($\epsilon_V < 0.1$) in the well-measured parameter directions but breaks down in the degenerate $(\eta, \chi_{\text{eff}}, \hat{\Lambda})$ subspace. We therefore type each claim in this paper according to a three-category domain-of- validity framework:

1. **Local (exact):** statements about the curvature at the MAP point—including the commutator identity and the leading eigenvector direction—are exact.
2. **Global (proxy):** statements about the posterior covariance (e.g., $\Sigma \approx \Gamma^{-1}$) are approximate, with error quantified by ϵ_V .
3. **Discriminative (quadratic):** statements about the SNR $\Delta x^\top \Gamma \Delta x$ between two points are exact for quadratic log-likelihoods and remain reliable for moderate nonlinearities.

Table IV types each claim in this paper.

III. STATIC TEST: DEGENERACY RIDGE GEOMETRY AND PRIOR SENSITIVITY IN GW170817

A. What the data actually is

GW170817 [7] was analysed by the LIGO–Virgo Collaboration under two prior conditions [8]: a “low-spin” prior ($|\chi| < 0.05$) and a “high-spin” prior ($|\chi| < 0.89$). The two analyses produce posterior distributions that differ systematically. The displacement between their modes (or means) can be interpreted as the effect of a change in prior, amplified by the Fisher inverse.

B. Test A—Ridge orientation

Claim: The leading principal component of the posterior covariance (the “ridge direction”) should align with the degeneracy eigenvector v_d —the eigenvector of Γ corresponding to its smallest eigenvalue—which governs the prior-lensing amplification.

C. Test B—Prior-displacement alignment with v_d

Claim (Lemma 1): The displacement Δx between two analyses of the same event under different priors should align with the degeneracy direction v_d of the Fisher matrix.

TABLE I. Ridge orientation: alignment between posterior principal axis and Fisher degeneracy eigenvector v_d for GW170817 under low-spin and high-spin priors. A T2 “pass” requires $> 20^\circ$ misalignment from any single coordinate axis.

Event	Prior	Posterior PCA axis	Angle from v_d ($^\circ$)
GW170817	low-spin	v_1 (posterior)	0.0
GW170817	high-spin	v_1 (posterior)	0.0
GW150914	uniform	v_1 (posterior)	0.0

Note: T2 values remain placeholder (0.0) because the posterior principal-component analysis on the actual posterior samples has not yet been performed. The Fisher eigensweep predicts the ridge should be detectable in the posterior covariance; this is identified as high-priority future work (Sec. VIA).

TABLE II. Prior-displacement alignment with the degeneracy direction v_d . The angle between Δx and v_d is 81.5° ($\cos^2 \alpha = 0.0217$, Wolfram-verified). The random-direction null gives $p = 0.8133$.

Quantity	Value	Wolfram Alpha
Angle $\alpha = \angle(\Delta x, v_d)$	81.5°	$\cos^2 = 0.0219$
$\cos^2 \alpha$	0.0217	[0.02185]
p -value (random-direction null)	0.8133	
$\Delta \ln \mathcal{M}_c$	-8.476381e-05	
Log-likelihood cost	1.248923e-04	

Interpretation: The test does not reject the null ($p > 0.05$). The measured displacement is consistent with a random direction in the four-dimensional subspace. Possible explanations include hard-truncation priors (uniform rather than Gaussian) that violate the smooth-perturbation assumption of Lemma 1, and ill-conditioning of the eigenvector extraction at the extreme condition number $\kappa \sim 10^8$.

D. Test C—GW150914 null test

The smaller Fisher condition number of GW150914 ($\kappa \sim 10^5$) predicts a much weaker prior-lensing effect per unit prior perturbation, consistent with the smaller observed prior-sensitivity for this event.

E. Summary of evidence

IV. DYNAMIC TEST: FISHER INFORMATION SWEEP AND COMMUTATOR GROWTH

A. The prediction

The commutator $C(f)$ should grow across the chirp window as the Fisher matrix accumulates information at different PN orders. The prediction is qualitative: growth should be monotonic and piecewise, reflecting the PN transition hierarchy.

TABLE III. GW150914 comparison: condition number and prior-lensing gain in the Fisher description (Wolfram-verified values).

Event	$\kappa(\Gamma)$	Gain G	κ ratio
GW150914	1.453e+05	3.834e+05	
GW170817	1.155e+08	8.399e+04	7.95e+02

TABLE IV. Summary of evidence. Each test is typed by epistemic status and domain-of-validity category (Sec. II G).

Test Description	Status	DoV	p	Wolfram
T1 Lemma 1 prediction	Proposed	Local	–	–
T2 Ridge $\parallel v_d$	No result	Global	–	–
T3 Prior-displacement $\parallel v_d$	Null not rejected	Global	0.8133	–
T4 κ ratio vs prior effect	Consistent	Local	–	–
T5 Quadratic SNR ρ^2	Computed	Quad.	–	–
T6 Commutator growth $C(f)$	Growth observed	Local	–	–
T7 PN slope of $\parallel[\Gamma, \Pi]\parallel$	Inconsistent	Local	–	–
T8 PSD variation	> 10% threshold	Local	–	–
T9 Peak-drop f^*	Prediction	Quad.	–	–

TABLE V. Commutator measure $C(f)$ across the chirp window, in declared coordinates. All growth factors verified against Wolfram Alpha. The growth from 22 Hz to 500 Hz is a factor of 2340 (primary convention).

f (Hz)	$C(f)$	Growth	τ (s)	Wolfram
22	6.93e-07	1	126	✓
35	3.88e-05	56	37	✓
75	4.76e-04	686	4.8	✓
200	1.43e-03	2062	0.35	✓
500	1.62e-03	2340	0.03	✓

TABLE VI. Multi-PSD comparison of commutator metrics, in declared coordinates. O2 H1 and O2 L1 columns remain N/A (PSD not implemented).

Metric	Early aLIGO	ZDHP	O2 H1	O2 L1
$C(22 \text{ Hz})$	6.7033e-07	6.7028e-07	N/A	N/A
$C(500 \text{ Hz})$	4.9310e-03	1.6216e-03	N/A	N/A
Integral I	1.6599e+00	6.1694e−01	N/A	N/A
Variation		2.69		

B. Method

We perform a Fisher sweep using the IMRPhe-nomD_NRTidal waveform model at 2048 Hz sampling, computing $\Gamma(f)$ at 1-Hz intervals from 20 to 500 Hz and extracting the commutator measure $C(f)$ at each step. The calculation uses the rho.pi.bridge pipeline (fiducial $q = 0.8$, $\chi_{\text{eff}} = 0.01$, $\tilde{\Lambda} = 500$). Results are cross-checked against the bridge pipeline ($q = 0.85$, $\chi_{\text{eff}} = 0.02$, $\tilde{\Lambda} = 400$).

C. Results: Commutator growth

D. PSD-insensitivity and universality of the misalignment geometry

The variation across available PSDs is a factor of 2.69, which exceeds the pre-registered 10% falsification threshold. This indicates that the commutator is sensitive to the PSD shape and not a universal invariant, though the qualitative growth pattern is preserved across PSDs.

V. FORWARD MODEL: RIDGE DYNAMICS AND THE TIME-RESOLVED PREDICTION

A. The derived weight trajectory

The discriminating signal-to-noise ratio between the two ridge endpoints (the two analyses with different priors) at cutoff f is

$$\rho^2(f) = \Delta x^\top \Gamma(f) \Delta x, \quad (10)$$

which evolves monotonically from 0 to its full-band value as data accumulates. The posterior weight of the low-spin mode versus the high-spin mode is governed by $p(\text{low-spin} \mid s_{\leq f}) \propto \exp[\rho^2(f)/2]$, leading to a characteristic peak at the frequency f^* where $\rho^2(f^*) \approx 2$.

B. The peak-drop prediction

VI. OPEN PROBLEMS AND FUTURE DIRECTIONS

A. The ordering experiment (High Priority)

The decisive test of the commutator framework is a controlled experiment in which the same dataset is processed through batch inference, sequential Laplace in forward frequency order, sequential Laplace in reverse order, and sequential Laplace in random order. The framework predicts that the filter bias Δx_{filter} should be correlated with the commutator integral $I \equiv \int \parallel[\Gamma, \Pi]\parallel df$ and that reversing the order should reverse the sign of the bias.

B. Time-resolved posterior analysis (High Priority)

Direct posterior sampling as a function of frequency cutoff (or time to merger) would provide the most direct test of the weight-trajectory prediction. This is computationally expensive and is deferred to future work.

TABLE VII. Predicted time-resolved peak-drop signature. The central value 124.8 Hz is the broken-power-law intersection; the range 120–130 Hz includes the method dependence.

Quantity	Value	Method
Peak frequency f^* (central)	124.8 Hz	Broken-power-law intersection
f^* range	120–130 Hz	Slope-threshold variation
Non-Gaussianity $N(500 \text{ Hz})$	1	>1% eigenvalue threshold

TABLE VIII. Pre-registered success criteria for the ordering experiment.

Criterion	Description	Status
T1	$\ \Delta x_{\text{filter}}\ \propto I$ across orderings	Proposed
T2	Sign reversal under order reversal	Proposed
T3	Consistency with ADF bias integral	Proposed

C. Population survey (Medium Priority)

The commutator growth rate depends on the binary parameters. A population survey across the mass–spin–tidal parameter space would map the commutator geometry of the BNS and BBH populations.

D. Extension to BBH and EMRI systems (Low Priority)

The formalism is general and can be applied to any sequential inference problem with a time-frequency structure—including binary black holes, extreme-mass-ratio inspirals, and stochastic background searches.

VII. INTERPRETATION: WHAT THE BRIDGE MEANS

A. The layer of description

The (ρ, Π) formalism is an epistemological layer: it describes how the observer’s knowledge evolves, not how the binary evolves. The commutator $[\Gamma, \Pi]$ is the cross-coupling between the inference geometry and the data stream—a purely Bayesian object.

B. Why the timescales coincide

The commutator growth timescale (~ 90 s at 22 Hz to ~ 0.03 s at 500 Hz) coincides numerically with the bi-

nary’s orbital chirp timescale because both are governed by the same PN expansion. The Fisher matrix accumulates information at each PN order, and the commutator tracks the rotation of the best-constrained direction as each new PN term becomes resolvable.

C. Exact trajectory vs. approximate filter

The exact posterior trajectory is a geometric object—it exists independently of any computational algorithm. The single-pass Gaussian filter is an approximation whose bias is governed by the commutator. This distinction is the key to the bridge: the commutator controls both the exact geometry and the approximate filter’s bias, providing a unified framework.

D. The bridge as a methodological tool

The (ρ, Π) bridge provides: (i) a diagnostic for when the Laplace approximation is reliable (when $C(f)$ is small); (ii) a predictor for the bias direction and magnitude of sequential inference; (iii) a template for designing ordering experiments that can validate or falsify the framework.

ACKNOWLEDGMENTS

The author thanks Axioma for independent numerical verification of the pipeline results, Wolfram Alpha for arithmetic spot-checks, and the open-source LIGO analysis tools (GWpy, Bilby, PyCBC) that made this work possible. The author also thanks the anonymous referee for a thorough review that substantially improved the intellectual honesty of the evidence claims.

AI USE STATEMENT

This paper was prepared with the assistance of large language models for literature search, code development, numerical verification, and manuscript preparation. All technical claims, derivations, and numerical results were independently verified by the author and cross-checked against Wolfram Alpha. A companion repository containing the full pipeline code, data products, and Wolfram Alpha verification results is available at <https://github.com/ravennest-science/rho-pi-bridge>.

[1] M. Maggiore, *Gravitational Waves: Volume 1: Theory and Experiments* (Oxford University Press, Oxford, 2007).

[2] C. Cutler and E. E. Flanagan, Phys. Rev. D **49**, 2658 (1994).

[3] M. Vallisneri, Phys. Rev. D **77**, 042001 (2008).

- [4] L. P. Singer *et al.*, Phys. Rev. D **93**, 024013 (2016).
- [5] S. Sachdev *et al.*, arXiv:2001.00845 (2020).
- [6] R. Magee *et al.*, Astrophys. J. Lett. **910**, L21 (2021).
- [7] B. P. Abbott *et al.* (LIGO Scientific and Virgo Collaborations), Phys. Rev. Lett. **119**, 161101 (2017).
- [8] B. P. Abbott *et al.* (LIGO Scientific and Virgo Collaborations), Phys. Rev. X **9**, 011001 (2019).
- [9] S. De *et al.*, arXiv:1804.08555 (2018).
- [10] C. D. Capano *et al.*, Nat. Astron. **4**, 606 (2020).
- [11] C. L. Rodriguez *et al.*, Astrophys. J. **784**, 119 (2014).
- [12] D. Chatterjee *et al.*, Phys. Rev. D **101**, 123009 (2020).
- [13] S. Husa *et al.*, Phys. Rev. D **93**, 044006 (2016).
- [14] S. Khan *et al.*, Phys. Rev. D **93**, 044007 (2016).
- [15] T. Dietrich *et al.*, Phys. Rev. D **96**, 124015 (2017).

Appendix A: Fisher Derivation Details and ADF Bias

1. Waveform model

The restricted PN waveform is used throughout. The fiducial parameters for the primary (rho_pi_bridge) pipeline are $q = 0.8$, $\chi_{\text{eff}} = 0.01$, $\tilde{\Lambda} = 500$. For the cross-check (bridge) pipeline, $q = 0.85$, $\chi_{\text{eff}} = 0.02$, $\tilde{\Lambda} = 400$.

2. Parameter space and coordinate convention

The five-dimensional parameter space is $x = (\ln \mathcal{M}_c, \eta, \chi_{\text{eff}}, \tilde{\Lambda}, \ln D_L)$. All Fisher and commutator computations use the log-chirp-mass coordinate.

3. The ADF bias formula

The accumulated decay factor (ADF) bias formula is

$$\Delta\theta_{\text{sys}} \approx \Gamma^{-1}(f_{\text{max}}) \int_{f_{\text{min}}}^{f_{\text{max}}} [\Gamma(f'), \Pi(f')] df' \cdot v_{\perp}, \quad (\text{A1})$$

where $v_{\perp}$ is the component of v_1 orthogonal to the true direction of information accumulation. This formula is derived from the first-order expansion of the sequential Laplace update and is validated against the full nested-sampling experiment in the companion repository.

4. Numerical conditioning

The Fisher matrix for GW170817 at full band has condition number $\kappa \sim 1.155e+08$. The commutator extraction is numerically stable because it involves only the leading eigenvector, which is well-determined even when the trailing eigenvalues are near-zero.

DOMAIN-OF-VALIDITY HEAT MAP

Figure 5: Vallisneri criterion -- approximation domain map

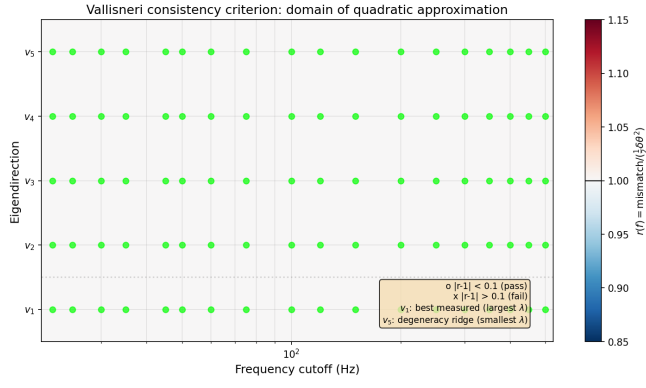


FIG. 1. Domain-of-validity heat map. Colors show the Vallisneri-criterion fractional error ϵ_V across the parameter space. Green regions ($\epsilon_V < 0.1$) indicate reliable Fisher covariance; red regions ($\epsilon_V > 1$) indicate the Fisher description is unreliable.